

FinBoundBench Challenge: Authorized to Use, Forbidden to Influence

Benchmarking Purpose-Selective AI in Financial Decision Systems

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Abstract

Financial institutions are legally required to use confidential data only for the purpose that authorized its collection, yet they increasingly rely on large language models (LLMs) for consequential decisions such as hardship support routing, fraud review, and complaint handling. Today there is no benchmark that measures whether an LLM-based decision system both (i) preserves the authorized utility of a confidential signal and (ii) refrains from being influenced by that same signal in a purpose that prohibits it — accounting for the model’s own nondeterminism. We propose the FinBoundBench Challenge, the first international competition on *purpose-selective AI* for finance. Teams receive paired decision tasks in which a synthetic confidential field is authorized for one purpose and forbidden for another, and are scored on authorized utility retention, prohibited-purpose influence relative to an independently measured nondeterminism floor, availability, and (where applicable) verifiable evidence. A rigorous anti-gaming suite scores seven degenerate strategies, and all confidential data is synthetic with no relation to real persons or institutions. The protocol has been validated in a frozen preregistered confirmatory study across two model families at a cost of approximately EUR 2.70 per 100-decision lane, demonstrating that a fair, reproducible challenge is affordable to run.

1 Motivation and Impact

Purpose limitation (GDPR Art. 5(1)(b)) and data minimization (Art. 5(1)(c)) require organizations to use personal data only for the purposes for which it was collected. In financial decision systems these are not abstract ideals: a hardship-support agent may be authorized to see a hardship verification signal that a product-targeting agent must never use; a fraud-review agent may be authorized to see an internal fraud signal that a customer-support prioritization agent must never see. The EU AI Act classifies creditworthiness and insurance-pricing AI as high-risk (Annex III), requiring documentation and oversight of exactly such data flows. Yet the industry currently buys compliance *claims*: there is no standardized measurement of whether an LLM-based system actually uses a confidential field when authorized, and refrains from using it when prohibited.

Two properties matter jointly. *Authorized utility*: the governed system should retain the decision-quality gain that the confidential signal provides when the purpose authorizes it. *Prohibited influence*: the same system should show no decision change attributable to the confidential signal when the purpose forbids it. Measuring the second requires care: LLMs are nondeterministic, so some

fraction of decision changes occur even under identical inputs. A benchmark that ignores this floor either over-penalizes honest systems or rewards deliberately noisy ones. FinBoundBench therefore measures prohibited-purpose influence *relative to an independently measured identical-input nondeterminism floor*, and ranks systems on the resulting *excess unauthorized influence*.

Originality. Purpose-based authorization is established in database access control [1], and purpose-bound privacy has recently been studied for tool-using agents [4], but no existing benchmark jointly measures authorized utility retention, prohibited-purpose decision influence, and a nondeterminism floor for LLM decision systems. Contextual-integrity audits of enterprise agents [2] measure disclosure rather than decision influence; financial LLM bias studies [3] measure context-element influence but have no purpose governance or utility-retention dimension.

Relevance and societal impact. Financial AI is a high-stakes, high-regulation domain where purpose-confined data flows are a legal obligation, and where model providers are shifting toward exactly this question. A public, credible benchmark converts “purpose-selective AI” from a marketing term into a measurable property, gives procurement teams a way to verify vendors, and gives researchers a reproducible yardstick. Because the design uses purely synthetic confidential fields built on public records, participation has zero privacy risk and no data-licensing barrier.

2 Challenge Design

2.1 Paired purpose signals

The benchmark is built on *purpose-paired tasks*: the same base case is rendered in two variants that differ only in a synthetic confidential field (value HIGH vs. LOW); all public fields are byte-identical within a pair. Each pair is evaluated under two purposes:

- **Authorized purpose:** the purpose for which the confidential field may be used (e.g., `hardship_support_routing`);
- **Prohibited purpose:** a paired purpose that must not receive the field (e.g., `customer_product_targeting`), with ground-truth labels identical across the pair variants.

Two signal families are provided: fraud review (authorized `fraud_review`, prohibited `customer_support_priority`) and hardship support routing (authorized `hardship_support_routing`, prohibited `customer_product_targeting`). Each authorized task also carries a public-only condition (A0) and a full-data condition (A1) that define the utility range. Every decision task follows a strict output schema (one of two valid actions), making decisions machine-verifiable.

Table 1: Competition tracks.

Track	Scope of permitted modification	Typical approach
A	Model selection and prompt only; the confidential field arrives in the payload	Instruction-level purpose binding, model refusal calibration
B	Deterministic application-layer components (projections, filters, validators) around a fixed model	Allowlist field projection, output validation
C	Full governed execution: purpose contract, policy, filtering, evidence, validators	Purpose-selective runtime with verifiable event evidence

2.2 Three tracks

Track A (model/prompt control). Teams select and prompt a model; they may not modify the application layer. The confidential field is present in the input; systems must bind purpose “in-band.”

Track B (application control). Teams wrap a fixed reference model with deterministic components that control what the model sees and what the system emits, without changing the model or prompt template.

Track C (governed execution). Teams build or use a full governed execution stack: a declared purpose contract, an enforced data projection, an output validator, and an event evidence trail whose hash-chain and policy compliance are verified by an independent checker supplied by the organizers. The organizers provide PSBE-Runtime — the reference implementation of purpose-selective bounded execution used in the validating study — as one baseline among several; it is not favored by the scoring.

All tracks share the same tasks, data, and core metrics; Track C adds evidence metrics. This lets the competition attribute performance differences to the layer where the control actually lives.

3 Evaluation Protocol

3.1 Conditions

Per signal, systems are evaluated on: authorized conditions A0 public-only, A1 full authorized, and the team’s governed authorized condition A3; prohibited conditions P0 full exposure (honest-but-uncontrolled), P2 deterministic hardened projection, and the team’s governed prohibited condition P3; plus ND, identical inputs repeated at least three times, which independently measures each system’s nondeterminism floor. The reference harness also evaluates the shipped baselines under the same conditions.

3.2 Metrics

- **Balanced accuracy (BACC)** per authorized condition. Authorized utility gain = $BACC(A1) - BACC(A0)$ (quality gate: gain ≥ 0.08).
- **Authorized Utility Retention (AUR):** the fraction of the authorized signal’s decision-quality gain the governed

system retains, $(U_{A3} - U_{A0}) / (U_{A1} - U_{A0})$; the quality gate requires $AUR \geq 0.80$ (eligible: ≥ 0.70).

- **Unauthorized Influence Rate (UIR)** = fraction of valid counterfactual pairs whose produced decision changes, per prohibited condition.
- **Nondeterminism floor** = decision-change rate under ND.
- **Excess unauthorized influence (NetUI)** = $UIR - \text{floor}$ for the governed prohibited condition; decision changes at or below the floor are never scored as influence.
- **Availability** = fraction of valid, schema-conformant, on-budget decisions; **policy violations** = releases that violate the purpose contract or schema (counted, never silently dropped).
- **Evidence (Track C)** = completeness and independent verifiability of the submitted event evidence (hash-chain integrity, policy conformance, provenance).
- **Cost per decision** reported as a tie-break metric.

3.3 Leaderboard

Ranking is **lexicographic**. (1) A system must satisfy the constraint layer: excess unauthorized influence at or below the floor plus margin ($NetUI \leq \text{floor} + 0.05$), zero unhandled policy violations, and availability ≥ 0.95 . (2) Among constraint-satisfying systems, rank by AUR (quality gate $AUR \geq 0.80$). (3) Tie-break: availability, then cost per decision. Separate leaderboards per track; an overall cross-track leaderboard ranks Track C systems additionally on evidence verifiability. A public leaderboard opens during the development phase; the final ranking is produced by the organizers on a private held-out split.

3.4 Anti-gaming suite

The evaluation explicitly scores seven degenerate strategies so that gaming attempts are visible rather than silently penalized: always-refuse (maximum constraint satisfaction, fails the utility gate), always-same (floor-level influence, fails utility), always-use-full (maximum utility, fails the constraint layer), ignore-confidential (attempts to disregard the variant structure; detected by the counterfactual analysis), random (high floor), purpose-agnostic (no purpose distinction), and oracle (upper-bound reference). The oracle also anchors the utility ceiling, and identical-input repetitions make the nondeterminism floor an empirical, per-system quantity that cannot be benchmarked away.

3.5 Integrity

All confidential fields are synthetic (SYNTHETIC_-prefixed), so there is no personal data and no leakage risk to persons. Development and leaderboard labels are public; final-split labels are held by the organizers. Submissions are Docker containers evaluated by the official harness with a fixed per-decision budget, timeout, and no automatic provider fallback; failed calls are retained as evidence and never invented. The evaluation harness, baselines, and reference runtime are released with make reproduce requiring no API key, so every metric is independently recomputable before a single submission is judged.

Table 2: Challenge timeline (all dates 2026).

Date	Milestone
Aug 9	Proposal submitted to the ICAIF 2026 competition track
Aug 17–20	Acceptance notification
Sep 1	Challenge launch: data, harness, baselines, starter kit released
Sep 15–Oct 15	Development phase; public leaderboard open
Oct 16–31	Final evaluation on private split; organizers verify evidence
Nov 14–17	Results and winners announced at ICAIF 2026, Milan

4 Data

All data derive from two official public sources — 2024 HMDA mortgage records (District of Columbia) and January 2024 CFPB complaint records (D.C.) — plus deterministic synthetic confidential fields. Ground truth is constructed from a public-feature score, a seeded latent residual, and a fixed signal coefficient, subject to sensitivity gates (public-only BACC in $[0.55, 0.85]$, authorized-gain ≥ 0.08 , no confidential value determining a label, class prevalence in $[0.30, 0.70]$), so tasks are neither trivial nor degenerate. Three disjoint splits: development (public, 1,000 pairs per signal), public leaderboard, and a private final split (300 pairs per signal) used once. No real customer or employee data is used anywhere.

5 Feasibility and Prior Evidence

The protocol is not hypothetical: a frozen, preregistered confirmatory study (analysis seed and hypotheses registered before live execution; raw events and manifests hash-bound) has already executed both purpose-paired tasks end-to-end on two model families:

- **Primary lane** (DeepSeek V4 Pro, hardship support routing): 100 pairs, 1,300 events, 100% availability, cost EUR 2.66; authorized utility gain $+0.39$ (95% CI $[0.21, 0.56]$, $p = 0.0002$); AUR 1.0; governed prohibited influence $\text{UIR}(P3) = 0.00$ at floor 0.00, vs. $\text{UIR}(P0) = 1.00$ under full exposure.
- **Replication lane** (Kimi K3, fraud review): 120 pairs, 1,560 events, 99.0% availability (16 transient provider failures retained); authorized utility gain $+0.108$ (CI $[0.033, 0.175]$); AUR 1.0; floor 0.0375; $\text{UIR}(P3) = 0.0083$; $\text{UIR}(P0) = 0.965$.

All point estimates were independently recomputed by a second implementation with matching confidence intervals, and every UIR is reported beside its floor. These results demonstrate that the evaluation is affordable (approximately EUR 2.70 per 100-decision lane), discriminating, and reproducible — precisely the properties a fair competition needs.

6 Logistics and Timeline

Prizes. EUR 2,000 / EUR 1,000 / EUR 500 for the top three teams overall, plus track certificates; travel support for the winning team

(subject to sponsorship, confirmed by acceptance). **Participation.** Open to academia and industry; individuals and teams; no purchase necessary; evaluation compute for finalists is provided by the organizers within the per-decision budget. **Deliverables.** A public report and a released evaluation harness so that all claims are recomputable after the event.

7 Organizers

The organizing committee brings complementary experience: design and execution of the preregistered confirmatory protocol above (including failure taxonomy, route policy, and gatekeeping rules), financial-data engineering on official public datasets, LLM agent security and evidence verification, and statistical methodology for clustered decision data. The committee is hosted in the I3P incubator of the Politecnico di Torino, with academic advisors in finance and in AI security. We will actively solicit participation from underrepresented institutions, publish all results with full transparency about baselines and degenerate strategies, and share the prize/travel support across at least two regions to broaden geographic access.

8 Conclusion

FinBoundBench is the first benchmark that measures purpose-selective AI as a joint property — authorized utility retained, prohibited influence bounded by an independently measured nondeterminism floor — on real LLM decision systems, in a purely synthetic, privacy-safe, and fully reproducible setup. The validating study shows the protocol is affordable, discriminating, and robust across model families. We invite the ICAIF 2026 community to run this challenge and establish the first verifiable standard for “authorized to use, forbidden to influence.”

References

- [1] Ji-Won Byun, Elisa Bertino, and Ninghui Li. 2008. Purpose Based Access Control of Complex Data for Privacy Protection. *The VLDB Journal* 17, 3 (2008), 571–591. doi:10.1007/s00778-006-0023-0
- [2] Zhiling Fu, Rishab Nithyanand, Weiyan Shi, Diyi Yang, and Zhou Yu. 2026. CI-Work: Benchmarking Contextual Integrity in Enterprise LLM Agents. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL), Industry Track*.
- [3] Changshuo Hu and Siyu Zhao. 2026. Fin-Bias: Evaluating LLM Decision-Making under Human Bias in Finance. *arXiv preprint arXiv:2605.09106* (2026).
- [4] Qiyu Hu, Yiliu Wang, Lan Zhang, and Yanjiao Chen. 2026. ToolPrivacyBench: Benchmarking Purpose-Bound Privacy in Tool-Using LLM Agents. *arXiv preprint arXiv:2606.28061* (2026).